B.TECH · MACHINE LEARNING PROJECT

Computer Science and Information Technology

Precision Farming Using Drone Imagery and CNN-Based Crop Analysis

Academic Year: 2026–27 | [College / University Name]

TEAM MEMBERS

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PROJECT GUIDE

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Introduction

Background

- Agriculture feeds the world and depends on timely crop monitoring
- Drones now give low-cost, high-resolution aerial views of whole fields
- Deep learning can read these images automatically

Domain Overview

- Precision farming: data-driven, site-specific crop management
- Computer vision on UAV imagery
- CNNs learn visual features; YOLO detects and localises objects

Why It Is Needed

- Manual field scouting is slow, costly and inconsistent
- Early problems spread before they are noticed
- Targeted action cuts wasted water, pesticide and labour

Idea: drone images in → CNN + YOLO analysis → crop-health problems located and classified.

Problem Statement

Existing Problem

- Crop monitoring is mostly done by walking the field
- It is time-consuming and needs significant manual labour
- Large fields cannot be inspected regularly

Limitations / Challenges

- Early disease, insect damage and stress are hard to spot by eye
- Results depend on the inspector’s experience
- Delayed detection leads to yield loss
- Aerial images need automated analysis at scale

Who Is Affected

- Farmers, especially small and medium holdings
- Agricultural officers and extension workers
- Consumers, through food supply and prices

Motivation

Why this project

Combines two fast-growing areas – drones and deep learning – in a practical, high-impact use case.

Real-world need

Farmers need quick, low-cost and reliable ways to spot crop problems before they spread.

Research / industry motivation

Precision agriculture and UAV-based crop monitoring are active research topics and a growing agri-tech industry.

Objectives

MAIN OBJECTIVE

Build a CNN-based system that analyses drone crop images and detects crop-health problems automatically.

1. Detect and locate Abiotic, Insect and Disease problems in drone images
2. Reduce manual inspection effort and labour
3. Enable early detection to support timely farming decisions
4. Present results through a simple web interface
5. Promote precision and sustainable agriculture

Existing System

Current Approach

- Manual field visits and visual inspection
- Expert opinion on suspected plants
- Periodic, sample-based checking

Existing Technologies

- Satellite imagery (coarse, weather-dependent)
- Basic image processing and vegetation indices
- Image-level classification models (healthy vs diseased only)

Limitations

- Slow, labour-intensive and subjective
- Low resolution or infrequent satellite data
- Classification gives no location of the problem
- Not suited to large fields

Proposed System

Proposed Solution

- Drones capture high-resolution crop images
- Images are preprocessed (resize, normalise, augment)
- A CNN-based YOLO model detects problems and draws bounding boxes
- Each detection is labelled Abiotic, Insect or Disease
- Farmers upload an image on a website and get the result

How it addresses the limitations

Slow manual scouting	▶	Automatic analysis in seconds
Hard to see early problems	▶	Learned visual features flag early signs
No location information	▶	Bounding boxes show where the problem is
Large fields	▶	Drone coverage + scalable software

System Architecture

INPUT

- Drone images of the crop field
- Uploaded through the website

PROCESSING

- Resize
- Normalise
- Augment

MODEL

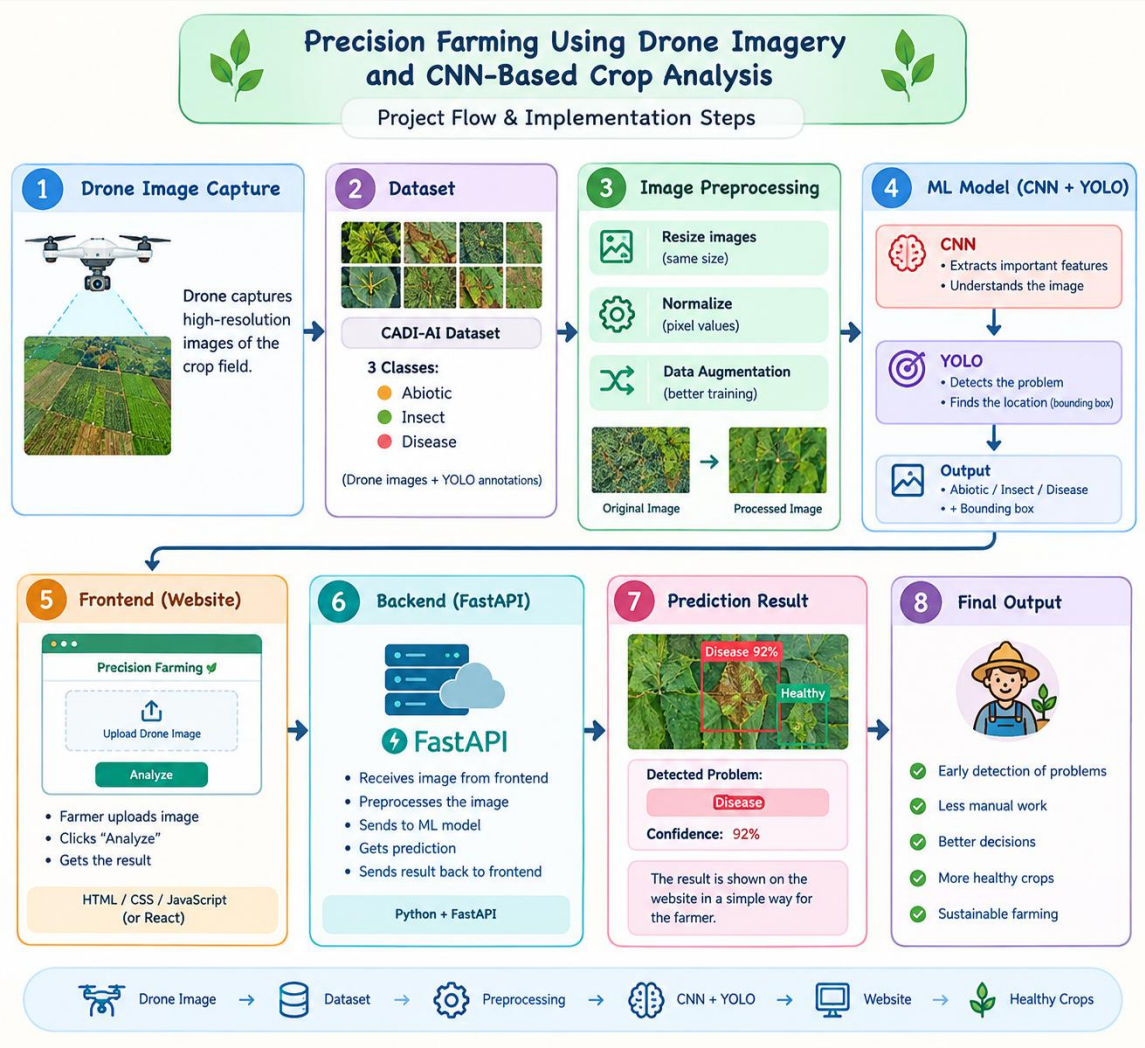
- CNN feature extraction
- YOLO detection
- Bounding boxes

OUTPUT

- Abiotic / Insect / Disease
- Bounding box + confidence
- Shown on result page

Web layer: Frontend (HTML / CSS / JavaScript) → FastAPI backend → trained model → prediction returned to the frontend.

Methodology / Workflow



1. Capture drone images
2. Collect CADI-AI dataset
3. Preprocess images
4. Train CNN + YOLO model
5. Build frontend + FastAPI backend
6. Predict and show result

Technology Stack

Languages

- Python
- HTML / CSS / JavaScript

Frameworks

- TensorFlow / Keras
- FastAPI

Libraries

- OpenCV
- NumPy · Pandas
- Matplotlib

Detection

- YOLO
- CNN backbone

Data

- CADI-AI (Hugging Face)
- YOLO annotation files

Tools / Hardware

- Jupyter / VS Code
- Drone camera; GPU (if used)

Database: image files and YOLO label files are stored in folders; no separate database is required.

System Modules / Components



User Interface

Upload drone image, click Analyze



Data Collection

CADI-AI drone images + labels



Preprocessing

Resize, normalise, augment



Core ML Module

CNN + YOLO detection model



Prediction Module

Class, bounding box, confidence



Visualization

Annotated image on result page



Backend API

FastAPI links UI and model



Storage

Dataset and trained weights

Implementation

1. Download CADI-AI and read the YOLO labels
2. Split into train / validation / test sets
3. Resize, normalise and augment images
4. Train the CNN + YOLO detector
5. Save the trained model weights
6. Expose a FastAPI /predict endpoint
7. Connect the upload page to the API
8. Show the annotated result to the user

Pipeline outline (pseudo-code)

```
data   = load_cadi_ai(split)
data   = resize(data); normalise(data)
data   = augment(data)
model  = build_cnn_yolo(classes=3)
model.train(data.train, data.val)

# FastAPI endpoint
@app.post("/predict")
def predict(image):
    boxes = model.detect(preprocess(image))
    return {"class": ..., "box": ...,
            "confidence": ...}
```

Dataset / Input Data — CADI-AI

Dataset Overview

NAME	CADI-AI
TYPE	Drone-based agricultural images
SOURCE	Hugging Face
ANNOTATIONS	YOLO format (class + box)
SPLIT	Training, validation, testing

Classes

- 0 — **Abiotic**
- 1 — **Insect**
- 2 — **Disease**

Preprocessing

Resize
Uniform input size

Normalise
Scale pixel values

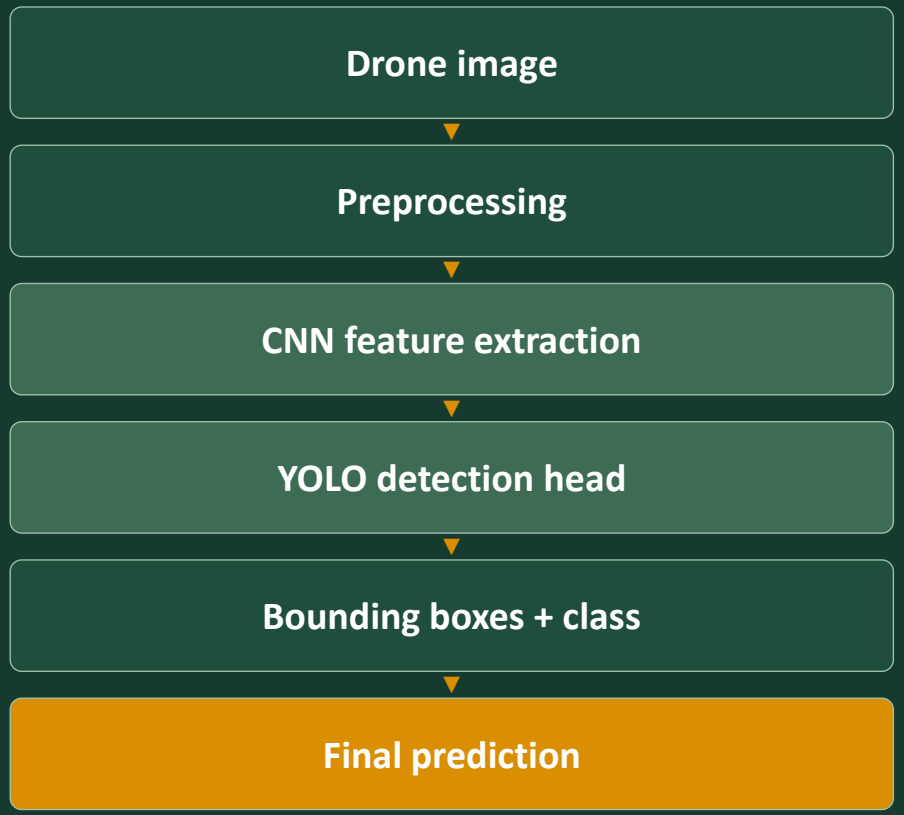
Augment
Flip, rotate, brightness

Labels
YOLO boxes kept in sync

CADI-AI is an object-detection dataset with three classes — not a simple Healthy / Diseased classification set. Image counts: [add from your split]

Algorithm / Model — CNN-Based YOLO

Model architecture



Why it was selected

- CNNs learn visual patterns from images
- YOLO detects and localises in one pass
- Fast enough for near real-time use
- Matches the annotated CADI-AI dataset

Key parameters

- Classes: 3
- Input size: [add]
- Epochs: [add]
- Batch size: [add]
- Learning rate: [add]
- Confidence threshold: [add]

Results

Accuracy / mAP

Precision

Recall

F1-score

Training / validation graphs

Insert loss curve, mAP curve and confusion matrix from your training run

Method	Precision	Recall	F1
Manual inspection	–	–	–
Image classification only	[add]	[add]	[add]
Proposed CNN + YOLO	[add]	[add]	[add]

Replace "—" and "[add]" with your final evaluation values.

Screenshots / Demo



Insert screenshot

Home page

Landing page of the website



Insert screenshot

Input screen

Upload drone image + Analyze



Insert screenshot

Processing screen

Model analysing the image



Insert screenshot

Result screen

Annotated image, class, confidence

Advantages

Faster monitoring

Whole fields analysed in minutes instead of days

Early detection

Problems spotted before they spread

Locates the problem

Bounding boxes show exactly where

Less labour

Reduces manual field inspection

Better decisions

Targeted use of water, pesticide and time

Easy to use

Simple upload-and-analyse website

Improvement over existing approach: automated, location-aware detection instead of manual or image-level checking.

Applications

Real-World Applications

- Crop disease and pest scouting
- Early warning for abiotic stress (water, nutrient, weather)
- Targeted spraying and irrigation planning
- Field health reports for agronomists
- Yield-loss reduction and crop insurance assessment

Target Users / Industries

- Farmers and farmer groups
- Agricultural departments and extension officers
- Agri-tech companies and drone service providers
- Research institutions
- Food supply and insurance sectors

Limitations

Current Limitations

- Detects only three classes: Abiotic, Insect, Disease
- No specific disease or pest names
- Works on uploaded images, not live drone video
- Performance depends on image quality and lighting

Dataset / Technical Constraints

- Single dataset (CADI-AI); limited crop and region variety
- Training needs a capable GPU
- Annotation quality limits accuracy
- Not yet tested across seasons or in real field trials

Future Scope

Improvements

Larger, more diverse datasets; fine-grained disease and pest classes

Additional features

Live drone video, severity estimation, treatment advice, multilingual UI

Deployment / scaling

Mobile app, cloud hosting, edge devices on the drone, GPS-based field maps

Conclusion

What was achieved	A CNN-based YOLO system that detects Abiotic, Insect and Disease problems in drone crop images, with a simple web interface.
Key outcome	Faster, earlier and less labour-intensive crop-health monitoring that supports better farming decisions.
Overall contribution	A practical step toward precision and sustainable agriculture using drones and deep learning.

Thank You